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Research Article

Fabrication and Excellent Performances of Bismuth Telluride-Based Thermoelectric Devices

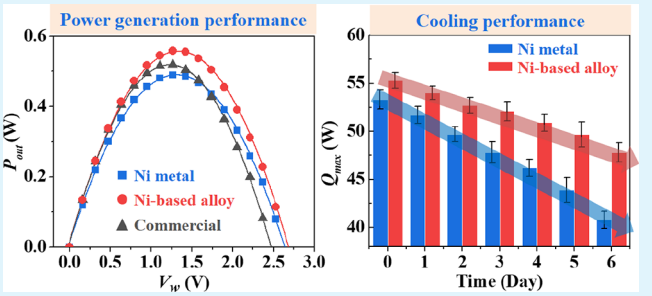
Wu Zhu, Ping Wei,* Jianqiang Zhang, Longzhou Li, Wanting Zhu, Xiaolei Nie, Xiahan Sang, Qingjie Zhang, and Wenyu Zhao*

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ABSTRACT: The barrier layer between thermoelectric (TE) legs and electrodes has crucial impact on the electrothermal conversion efficiency of the TE device; however, the interfacial reaction of the Ni metal barrier layer with TE legs in traditional Bi_2Te_3-based devices is harmful to the device performance. Herein, a high-quality barrier layer of a Ni-based alloy has been fabricated on both n-type and p-type Bi_2Te_3-based TE legs by the electroplating method. The in situ XRD results indicate that the as-prepared Bi_2Te_3-based TE legs with a Ni-based alloy barrier layer remain stable even at 300 °C. The high-resolution high-angle annular dark field scanning transmission electron microscopy images reveal that the Ni-based alloy barrier layer has more excellent stability than that of the Ni metal barrier layer. The Bi_2Te_3-based TE devices with excellent structural and performance stabilities were assembled with the as-grown high-performance n-type and p-type Bi_2Te_3-based leg with a Ni-based alloy barrier layer, which have lower internal resistance and higher cooling and power generation performances. A maximum cooling temperature difference over 65 K and a maximum cooling capacity of 55 W were obtained for the high-performance Bi_2Te_3-based TE devices. This work provides a new strategy for high-temperature applications of commercial Bi_2Te_3-based TE devices.



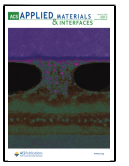
KEYWORDS: *thermoelectric device, bismuth telluride, barrier layer, interfacial reaction, thermal stability*

1. INTRODUCTION

Thermoelectric (TE) devices can realize all-solid-state energy conversion between electricity and heat based on the Seebeck effect and Peltier effect, having important applications in waste heat recovery, radioisotope TE generation, on-chip cooling, and high-accuracy temperature control.[1,2] TE devices have a compact structure and are composed of several pairs of TE legs and two copper-clad ceramic sheets, therefore having the advantages of small size, no noise, zero pollution, and no moving parts.[3,4] The TE materials used in the commercial devices are still bismuth tellurides including n-type and p-type Bi_2Te_3-based alloys.[5−7] During the manufacturing process of Bi_2Te_3-based TE devices, a crucial procedure is sandwiching the n-type and p-type Bi_2Te_3-based legs in between two ceramic sheets through hot-pressing soldering, where n-type and p-type Bi_2Te_3-based legs are alternately connected in series through the Cu strip electrodes cladded on the ceramic sheets.[8−10] However, it is very challenging to combine Bi_2Te_3-based legs with the Cu electrodes in the soldering process. Bi_2Te_3-based alloys are not well wet with Sn-based solders, and it is therefore difficult to ensure a strong bonding strength:[11,12] a weak bonding probably causes electrical and thermal disconnections or impaired mechanical strength. The Sn-based solder will react with Bi_2Te_3-based alloys to form the SnTe alloy, which results in the increase in the interfacial electrical and thermal resistance, a decrease in the mechanical strength, and even solder failure.[13,14] To solve this problem, an effective way is preplating a barrier layer on the surface of Bi_2Te_3-based TE legs.[15−17] It is expected that the barrier layer can not only strengthen the bonding strength between Bi_2Te_3-based TE legs and Cu electrodes but also suppress the above chemical reaction between the TE legs and solder.[17,18] Therefore, the barrier layer plays a very important role in the manufacturing process of the Bi_2Te_3-based TE devices.

In commercial Bi_2Te_3-based TE devices, metallic Ni is widely used as the barrier layer due to the advantages as follows: (1) Ni has a thermal expansion coefficient similar to that of Bi_2Te_3-based alloys. The good thermal expansion match at the interface may ensure the good thermal stability during the heating and cooling cycles; (2) the small lattice mismatch between Ni and Bi_2Te_3-based legs may ensure a good contact

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with less transverse interior stress;[12,19] (3) the electrical contact between Ni and Bi_2Te_3-based legs is Ohmic, and hence, the contact electrical resistance is relatively small (about 4 $\mu\Omega$ cm^2);[20] (4) the manufacturing cost of the Ni metal barrier layer is low. However, preplating the Ni metal barrier layer is actually not an ideal choice: when the Bi_2Te_3-based TE device serves beyond 200 °C, the Ni metal barrier layer becomes unstable and easily reacts with Bi_2Te_3-based legs to form the interfacial reaction layer (IRL) of NiTe.[15] The formation of the IRL is not only harmful to the TE performance of Bi_2Te_3-based TE materials due to the doping effect but also causes a significant increase in the interfacial electrical resistance between the Cu electrode and Bi_2Te_3-based TE legs. A p-type $(Bi_{1-x}Ni_x)_2(Te_{2.7}Se_{0.3})_{3-\delta}$ IRL was found at the interface between the n-type $Bi_2(Se,Te)_3$ TE legs and Ni metal barrier layer.[19] The presence of the p-type IRL formed on the n-type Bi_2Te_3-based TE legs may cause a very high contact electrical resistance, which severely deteriorates the device performance. Therefore, fabricating a high-stability barrier layer is very important to overcome the interfacial reaction and improve the device performance.

Herein, a high-quality barrier layer of the Ni-based alloy was fabricated on both the as-grown n-type and p-type Bi_2Te_3-based TE legs by an electroplating method. The thermal stabilities of the as-prepared Bi_2Te_3-based TE legs with the Ni metal and Ni-based alloy barrier layers were investigated using the in situ XRD method. Two kinds of Bi_2Te_3-based TE legs with different barrier layers were annealed at 200 °C for different days to obtain the evolution of the IRL of the barrier layers with TE legs. The high-resolution high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) images reveal that the Ni-based alloy barrier layer has more excellent stability than that of the Ni metal barrier layer. The Bi_2Te_3-based TE devices with excellent structural and performance stabilities were assembled with the as-grown high-performance n-type and p-type Bi_2Te_3-based leg with the Ni-based alloy barrier layer, which have lower internal resistance and higher cooling and power generation performances. This work provides a new strategy for high-temperature applications of commercial Bi_2Te_3-based TE devices.

2. EXPERIMENTAL PROCEDURES

2.1. TE Material Preparation. By using the traditional zone-melting method, n-type and p-type Bi_2Te_3-based alloy rods were prepared. The sample preparation process includes (1) weighing high-purity elements (Bi, Te, Sb, and Se ingots, 4N purity, Wuhan Xinrong Co., Ltd.) according to the stoichiometric ratio of $Bi_2Te_{2.7}Se_{0.3}$ (n-type rods) and $Bi_{0.5}Sb_{1.5}Te_3$ (p-type rods), (2) sealing the weighed elements in conical-bottom quartz tubes under vacuum and placing them in a rocking tube furnace for a uniform melting, and (3) growing Bi_2Te_3-based alloy rods by melting the quartz tubes in a zone-melting furnace with an optimized speed. Several commercial Bi_2Te_3-based zone-melted ingots including the n-type and p-type ones were purchased from Hubei Saigre New Energy Technology Co., Ltd. for TE property comparison.

2.2. Barrier Layer Electroplating. The Ni-based alloy barrier layer was deposited on the n-type and p-type Bi_2Te_3-based legs by a chemical electroplating process. The electroplating solutions used for depositing the Ni-based alloy barrier layer were prepared with appropriate contents of alloying elements, sulfates, 5 g L^{-1} $NaCl_2$, 35 g L^{-1} H_3BO_3, 0.1 g L^{-1} $C_{12}H_{25}SO_4Na$, and 1 g L^{-1} saccharin. High-purity nickel sheets were used as the anode. The cathodes used were p-type $Bi_{0.5}Sb_{1.5}Te_3$ and n-type $Bi_2Te_{2.7}Se_{0.3}$ wafers with a diameter of 30 mm and a thickness of 1.6 mm, which were cut from zone-melted n-type and p-type Bi_2Te_3-based ingots using an internal circular slicer. A constant DC power supply was used to control the current density during the electroplating process. The technique conditions including current density, electroplating time, and solution temperature were optimized, which were 0.8 A dm^{-2}, 15 min, and 40 °C, respectively.

2.3. Device Fabrication. In order to evaluate the thermal stability of the interfaces between the Bi_2Te_3-based TE wafers and Ni-based alloy barrier layers, the electroplated TE wafers were sealed in quartz tubes under vacuum and placed in a muffle furnace for an isothermal heat treatment at 200 °C for different times. In order to further evaluate the effect of Ni-based alloy barrier layers on the thermal stability of the TE device, several plate-type Bi_2Te_3-based TE devices were fabricated according to the commercial TEC1-12706 specification. The assembly steps were similar to the industrialized procedure: first, the electroplated n-type and p-type Bi_2Te_3-based wafers were diced into TE legs with a size of 1.4 mm × 1.4 mm × 1.6 mm by a precision dicing machine. Second, 127 pairs of n-type and p-type Bi_2Te_3-based TE legs were arrayed in a rubber mold using a vibrating device and hot-welded within two 40 mm × 40 mm copper-clad ceramic sheets, where cladded strip copper electrodes were coated with a Sn-based solder paste. Finally, plate-type Bi_2Te_3-based TE devices with a size of 40 mm × 40 mm × 3.8 mm were assembled, followed by welding two conduction wires on and sealing them with a sealant. For comparison, traditional Ni metal barrier layers were also electroplated on p-type and n-type Bi_2Te_3-based wafers using the same electroplating process and the corresponding Bi_2Te_3-based TE devices with a Ni metal barrier layer were also fabricated by following the same assembly steps. Several commercial TEC1-12706 TE cooling devices were purchased from Hangzhou Ferrotec Co., Ltd. and Hubei Saigre New Energy Technology Co., Ltd. for performance comparison.

2.4. Structural Characterizations. An X-ray diffractometer (Rigaku, SmartLab) using Cu Kα radiation (λ = 0.154186 nm) was employed to in situ-analyze the phase compositions of the electroplated wafers and Bi_2Te_3-based TE materials. During the in situ test, a temperature gradient of 50 K and a holding time of 30 min were set. The surface morphology of the barrier layers and Bi_2Te_3-based TE materials was observed by field-emission scanning electron microscopy (FESEM, JEOL, JEM-7500F). Transmission electron microscopy (TEM, Thermo Fisher Scientific, Talos F200s) was used to characterize the inner microstructure and element distribution near the interfaces between the barrier layer and Bi_2Te_3-based TE legs. Focused ion beam (FIB, Thermo Fisher Scientific, Helios NanoLab UC) was used to prepare the TEM samples with a Ga ion-beam source and a lift-out system. A tensile strength tester was used to measure the bonding strength between the barrier layers and TE materials. Before the test, the electroplated wafers were cut into small squares with a typical dimension of 5 mm × 5 mm × 1.6 mm. After soldering the upper and lower sides of the wafers onto two copper blocks with a Sn-based solder, the copper blocks were clamped on the tensile strength tester. By stretching the two copper blocks, the bonding strength between the barrier layer and Bi_2Te_3-based material could be measured after tearing the electroplated barrier layer from the wafers. The electrical conductivity (σ) and Seebeck coefficient (α) were measured by simultaneously using a standard four-probe method on a commercial system (Cryoall, CTA-3/500). A laser flash method (NETZSCH, LFA-467) was used to determine the thermal diffusivity λ. The density ρ was measured by the Archimedes method, and the specific heat C_p was obtained using a differential scanning calorimeter (TA Q20). The thermal conductivity κ was calculated using $\kappa = \lambda \cdot \rho \cdot C_p$.

2.5. Device Performance Measurements. A KA6288-type semiconductor component test system (Huizhou Xinjun Electronic Technology Co. Ltd.) was employed to evaluate the cooling performance of Bi_2Te_3-based TE devices with the Harman method. The power generation performances of Bi_2Te_3-based TE devices were evaluated using self-made measuring equipment, as reported in another work.[21] During the power generation measurements, the hot-end temperature (T_h) and cold-end temperature (T_c) were controlled using a constant-temperature heater and a water cooling system, respectively. The temperature difference (ΔT) could be established in

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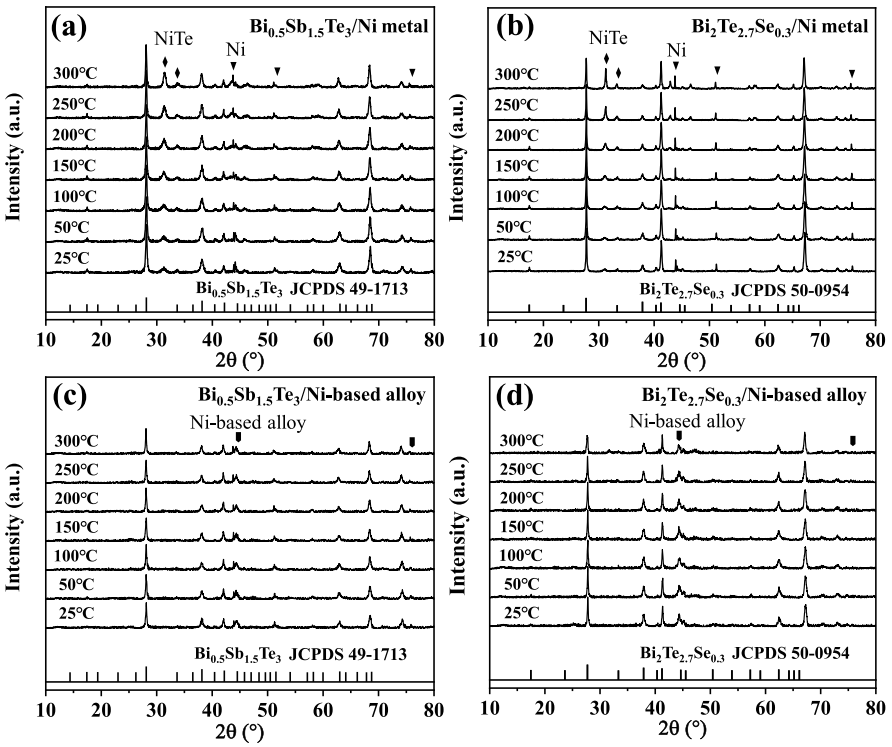


Figure 1. In situ XRD patterns. (a) p-type and (b) n-type Bi_2Te_3-based wafers with the Ni metal barrier layer; (c) p-type and (d) n-type Bi_2Te_3-based wafers with the Ni-based alloy barrier layer.

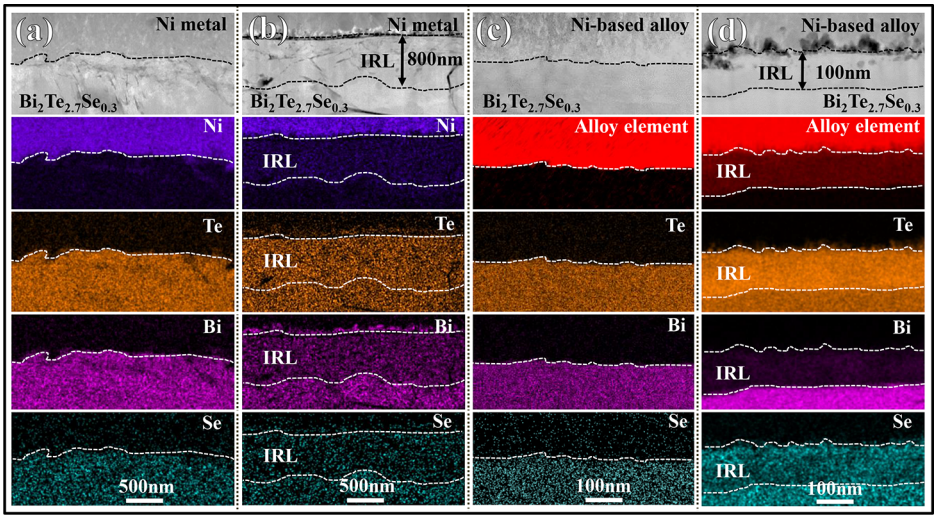


Figure 2. HAADF-STEM images and elemental mapping of the interfaces between the barrier layers and n-type Bi_2Te_3-based wafers. (a) As-prepared Ni metal barrier layer, (b) Ni metal barrier layer after having been annealed at 200 °C for 48 h, (c) as-prepared Ni-based alloy barrier layer, and (d) Ni-based alloy barrier layer after having been annealed at 200 °C for 48 h.

the range of 0−100 K. The working voltages (V_w) and currents (I_w) were measured using a SourceMeter (SMU, Keithley 2401). The open-circuit voltage (U_x) was obtained when $I_w = 0$. The output power (P_{out}) was calculated with $P_{out} = V_w \cdot I_w$.

3. RESULTS AND DISCUSSION

3.1. In Situ XRD of the Ni-Based Alloy Barrier Layer. In order to compare the thermal stability between the Ni-based alloy barrier layer and Ni metal barrier layer, the Bi_2Te_3-based wafers electroplated with two kinds of barrier layers were subjected to investigate with in situ XRD analysis during the heating process from 25 to 300 °C. Figure 1a,b shows the in situ XRD patterns of p-type and n-type Bi_2Te_3-based wafers with a Ni metal barrier layer. Before in situ heating, weak diffraction peaks of NiTe were observed in the XRD patterns of p-type and n-type Bi_2Te_3-based wafers with the as-prepared Ni metal barrier layer. This suggests that a small amount of NiTe was formed during the electroplating process. During the in situ heating process, the strength of the characteristic diffraction peaks of NiTe gradually increases as the temperature increases. This indicates that the Ni metal barrier layer easily reacts with Te from Bi_2Te_3-based wafers. Comparing with the cases of p-type Bi_2Te_3-based wafers, it can be found

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that the Ni metal barrier layer becomes more unstable above 150 °C and reacts severely with the n-type Bi_2Te_3-based wafer.

Figure 1c,d shows the in situ XRD patterns of p-type and n-type Bi_2Te_3-based wafers with the Ni-based alloy barrier layer. Before in situ heating, no diffraction peak of NiTe was observed in the XRD patterns of p-type and n-type Bi_2Te_3-based wafers with the as-prepared Ni-based alloy barrier layer. This suggests that the Bi_2Te_3-based wafers were stable during the electroplating process. During the in situ heating process, both p-type and n-type Bi_2Te_3-based wafers with the Ni-based alloy barrier layers still remained stable even when being heated to 300 °C. Their in situ XRD patterns only showed the diffraction peaks of the Bi_2Te_3-based compounds and Ni-based alloy.

The lattice parameters of the Bi_2Te_3-based alloys during the in situ heating are given in Table S2. Upon increasing the temperature, the size of *a* for all the Bi_2Te_3-based alloys slightly increased due to the thermal expansion effect. However, a different change behavior was observed in the size of *c*. The Bi_2Te_3-based alloys coated with the Ni-metal barrier layer exhibited a significantly increased *c* when the heating temperature exceeded 150 °C, while the Bi_2Te_3-based alloys coated with the Ni-based alloy barrier layer showed a slightly increased *c* value in the whole temperature range from 25 to 300 °C. This suggests that the Ni-based alloy barrier layer has much better thermal stability. The increased *c* should be attributed to the intercalation of Ni inside the van der Waals gap of the Bi_2Te_3-based alloys. Therefore, it can be concluded that the Ni-based alloy barrier layer may significantly improve the thermal stability of the Bi_2Te_3-based legs in the TE devices.

3.2. Microstructure of the Ni-Based Alloy Barrier Layer. The SEM photographs of the surface and cross-section of the n-type Bi_2Te_3-based wafer with the Ni-based alloy barrier layer (Figure S1) show that the thickness of the as-prepared Ni-based alloy barrier layer is around 1.5 μm, and there are no obvious pinholes on the surface. The tensile test results show that the bonding strength between the as-prepared Ni-based alloy barrier layer and Bi_2Te_3-based wafer was about 8 MPa, which is comparable with the wafer of the as-prepared Ni metal barrier layer (~8.5 MPa).

Figure 2 shows the HAADF-STEM images and elemental mapping of the interfaces between two kinds of barrier layers and n-type Bi_2Te_3-based wafers, where Figure 2a shows the as-prepared Ni metal barrier layer, Figure 2b shows the Ni metal barrier layer after having been annealed at 200 °C for 48 h, Figure 2c shows the as-prepared Ni-based alloy barrier layer, and Figure 2d shows the Ni-based alloy barrier layer after having been annealed at 200 °C for 48 h. It can be seen that both Ni metal and Ni-based alloy barrier layers compactly make contact with the n-type Bi_2Te_3-based wafers, which thus ensures a large bonding strength and good electrical conduction at the interfaces. The selected-area electron diffraction (SAED) patterns of the Ni metal barrier layer (Figure S2a) and Ni-based alloy barrier layer (Figure S2b) exhibit ring-shaped diffraction patterns, manifesting the polycrystalline features of the Ni metal and Ni-based alloy barrier layers. As shown in Figure 2a,c, the elemental mapping of Ni, Te, Bi, Se, and alloy elements suggests that all elemental distributions are uniform in the two kinds of the as-prepared barrier layers. These results show that the optimized electroplating conditions of the Ni metal barrier layer are also available for the Ni-based alloy barrier layer. During the electroplating process, Te diffusion at the interface between the Ni metal barrier layer and n-type Bi_2Te_3-based wafers forms NiTe. However, the elemental diffusion of Te becomes undetectable at the interface between the Ni-based alloy barrier layer and n-type Bi_2Te_3-based wafers, which is consistent with the XRD results (Figure 1). Elemental diffusion in the case of p-type Bi_2Te_3-based wafers is similar to that of n-type Bi_2Te_3-based wafers (Figure S3).

Besides the electroplating process, Te diffusion might take place during the operation of the TE device. In order to evaluate the thermal stability of the two kinds of barrier layers, both n-type and p-type Bi_2Te_3-based wafers electroplated with the Ni metal barrier layer and Ni-based alloy barrier layer had been annealed at 200 °C for 48 h in a vacuum environment. As shown in Figure 2b, the obvious elemental diffusion between Ni and n-type Bi_2Te_3-based wafers was clearly observed, which is in good agreement with the reported results.[12,19] It can be seen that in the case of the Ni metal barrier layer, the thickness of the IRL exceeds 800 nm for the n-type Bi_2Te_3-based wafers (Figure 2b) and reaches 500 nm for the p-type Bi_2Te_3-based wafers (Figure S3a). Especially, the IRL between the Ni metal barrier layer and n-type Bi_2Te_3-based wafers is not uniform, indicating that the IRL was difficult to control. In contrast, the formation of the IRL may be largely suppressed due to the presence of the Ni-based alloy barrier layer. As shown in Figure 2d, the thickness of the IRL is only 100 nm in the n-type Bi_2Te_3-based wafers with the Ni-based alloy barrier layer. It is noted that the IRL is much uniform and much thinner. The diffusion of Te and alloy elements was not obvious at the interfaces. For the p-type Bi_2Te_3-based wafer, the IRL is even thinner and its thickness is only 50 nm (Figure S3b). There is only a slight chemical reaction between the Ni-based alloy barrier layer and Bi_2Te_3-based wafers after having been annealed at 200 °C for 48 h. The driving force that results in the elemental diffusion at the interfaces between the TE materials and barrier layer should be closely related to the interfacial formation energy. The Ni-based alloy contains Co, which should play an important role in improving the stability. It has been reported that the Co/Bi_2Te_3 interface has a much lower formation energy than the Ni/Bi_2Te_3 interface,[22] which suggests that the Co/Bi_2Te_3 interface is more stable and not prone to the reaction as compared with the Ni/Bi_2Te_3 interface. Therefore, the use of the Ni-based alloy barrier layer is not only effective in hindering the Te diffusion and the formation of NiTe during the electroplating process but also provides an effective way to improve the thermal stability and service life of the Bi_2Te_3-based device.

3.3. Performance of the Bi_2Te_3-Based TE Device. To further reveal the beneficial effect of the Ni-based alloy barrier layer on the device performance, several standard TEC1-12706 devices were fabricated with the as-grown n-type and p-type Bi_2Te_3-based ingots. The TE properties of both n-type and p-type Bi_2Te_3-based ingots were measured, and the results are plotted in Figure 3. It can be seen that the TE properties of the as-grown Bi_2Te_3-based ingots exhibit similar temperature dependence as those of the commercial ingots. Their electrical conductivity gradually decreases upon increasing the temperature, manifesting a heavily doped semiconductor conduction behavior. In the temperature range of 300−500 K, the absolute Seebeck coefficient of all the as-grown ingots first increases and then decreases above 388 K, which stands for a turnover from extrinsic conduction to intrinsic conduction, where the bipolar effect dominates at higher temperatures.

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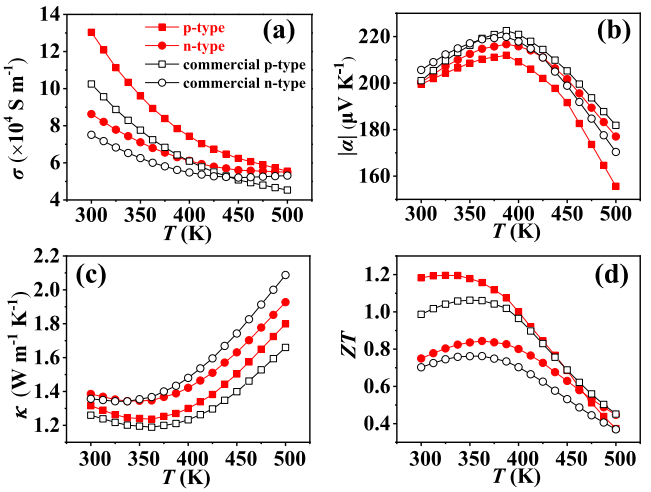


Figure 3. Temperature dependence of TE properties of the n-type and p-type Bi_2Te_3-based ingots. (a) Electrical conductivity σ, (b) absolute Seebeck coefficient α, (c) thermal conductivity κ, and (d) dimensionless figure of merit ZT.

Owing to the bipolar effect, the thermal conductivity increases dramatically above 360 K, which leads to decreased figure of merit ZT ($ZT = \sigma\alpha^2/\kappa$) at high temperature. Compared with the TE properties of commercial Bi_2Te_3-based ingots, the as-grown p-type and n-type ingots in this work have a higher electrical conductivity of 1.30×10^5 S·m^{-1} and 8.63×10^4 S·m^{-1}. Therefore, the TE devices assembled with the as-grown ingots should have a low internal resistivity if the additional interfacial electrical resistivity was ignored. The absolute Seebeck coefficient of the as-grown ingots is slightly lower than those of commercial ingots but exceeds 180 μV·K^{-1} in a wide temperature range of 300−470 K, which may ensure a large open voltage for power generation. Note that the difference in the thermal conductivities between the commercial p-type and n-type ingots is large and increases from 0.10 W·m^{-1}·K^{-1} at 300 K to 0.43 W·m^{-1}·K^{-1} at 500 K. However, the difference in the thermal conductivities between the as-grown p-type and n-type ingots subtly changes and exhibits a small variation around 0.07−0.11 W·m^{-1}·K^{-1}. The smaller difference of the thermal conductivities between the p-type and n-type legs in a TE device is very important for thermal transport, which may ensure a uniform heat flow across the device. Benefiting from the higher electrical conductivity and the lower lattice thermal conductivity (Figure S5), the as-grown Bi_2Te_3-based ingots possess higher ZT values as compared with those of the commercial ones. The ZT values at 300 K reach 1.18 and 0.75 for the as-grown p-type and n-type Bi_2Te_3-based ingots, respectively, and exhibit superior TE performance in a large temperature range of 300−470 K.

In order to explain the outperformed TE performance of the as-grown Bi_2Te_3-based ingots, Hall effect measurements were carried out at 300 K for the as-grown and commercial Bi_2Te_3-based ingots. The Hall effect measurement results are given in Table S1. It can be seen that the as-grown p-type Bi_2Te_3-based ingot exhibits a much higher carrier mobility (over 40%), which might stem from the better crystallization. The as-grown n-type Bi_2Te_3-based ingot also exhibits a higher carrier mobility, although the carrier concentration is higher, which also should be related to the better crystallization of the ingots. It is possible that the as-grown ingots were melted from the elements with higher purity, which ensures better crystallization.

In this work, the whole fabrication process of the Bi_2Te_3-based TE devices was performed by starting from the melting of elemental metals. The device fabrication steps are shown in Figure 4, which include (1) zone-melting the elemental metals to obtain the as-grown n-type and p-type Bi_2Te_3-based ingots, (2) cutting ingots to obtain the n-type and p-type Bi_2Te_3-based wafers, (3) electroplating the wafers to deposit the Ni metal barrier layer and Ni-based alloy barrier layer on the wafers, (4) dicing the electroplated wafers to obtain n-type and p-type Bi_2Te_3-based grains, (5) arraying the n-type and p-type Bi_2Te_3-based legs, and (6) assembling the devices. This fabrication process can also be used to assemble other devices with different sizes.

The internal resistance, maximum cooling temperature difference, and maximum cooling power of the as-assembled Bi_2Te_3-based devices were measured, and the results are listed in Table 1. It can be found that the as-assembled devices using the Ni-based alloy as the barrier layer have superior cooling performance. The as-assembled device with the Ni metal barrier layer exhibits an internal resistance of 2.10 Ω, a

Table 1. Internal Resistance R, Maximum Cooling Temperature Difference ΔT_{max}, and Maximum Cooling Power Q_{max} of Several TEC1-12706 Devices

device (barrier layer)	R (Ω)	ΔT_{max} (K)	Q_{max} (W)
device 1 (Ni-based alloy)	2.02	65.2	55.3
device 2 (Ni-based alloy)	2.05	64.5	54.8
device 3 (Ni-based alloy)	2.07	64.5	54.6
device 4 (Ni metal)	2.10	63.2	53.3
commercial 1 (Ni metal)	2.07	64	54
commercial 2 (Ni metal)	2.12	62.8	52.5

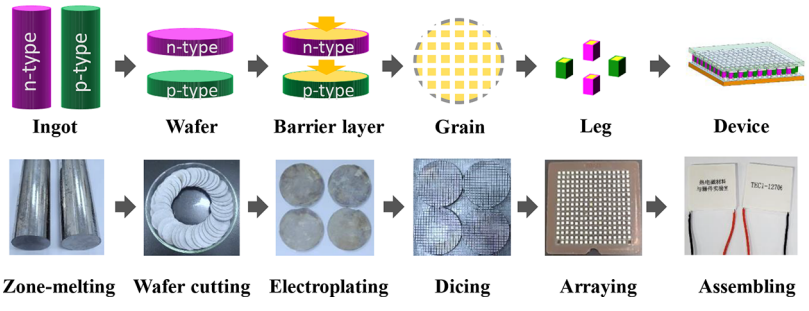


Figure 4. Manufacturing processes of the Bi_2Te_3-based devices with TEC1-12706.

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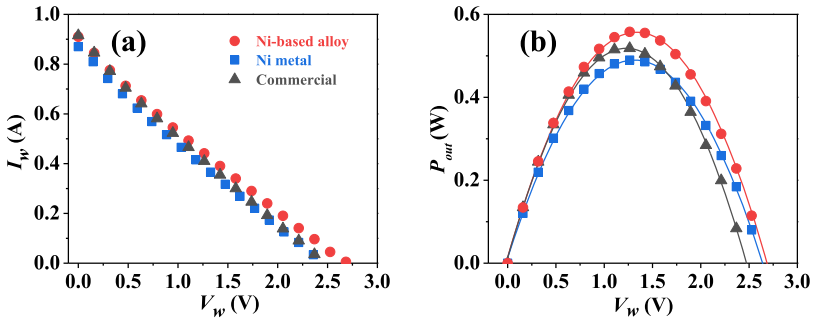


Figure 5. Power generation performances of the TEC1-12706 devices with different barrier layers. (a) Working current and (b) output power as a function of the working voltage. The hot-end and cold-end temperatures are at 364 and 300 K, respectively.

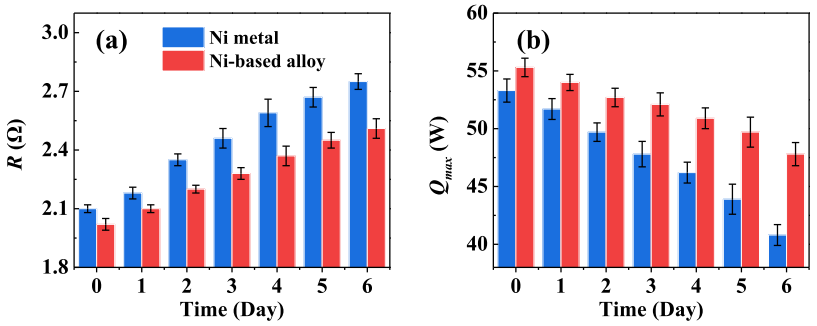


Figure 6. Effect of the barrier layer on the performance of the Bi_2Te_3-based TE devices. (a) Internal resistance R and (b) maximum cooling capacity Q_{max} of two kinds of Bi_2Te_3-based TE devices, which are fabricated by using Bi_2Te_3-based TE legs with the Ni metal barrier layer and Ni-based alloy barrier layer after having been annealed at 200 °C for different days.

maximum cooling temperature difference of 63.2 K, and a maximum cooling power of 53.3 W. However, these as-assembled devices with the Ni-based alloy barrier layer exhibit an internal resistance as low as 2.02 Ω, a larger cooling temperature difference exceeding 64.5 K, and a higher maximum cooling power of 55.3 W. The lower internal resistance in the case of the Ni-based alloy barrier layer should be due to the thinner IRL since the formation of IRLs between the barrier layer and Bi_2Te_3-based alloys leads to improved internal resistance.[19] Compared with commercial devices, our self-made devices with the Ni-based alloy barrier layer also possess better cooling performance. Therefore, the TE devices with the Ni-based alloy barrier layer have superior cooling performance that could be competitive with commercial products.

Since the TE devices have a maximum cooling temperature around 64 K, we performed the power generation test with the same temperature difference. During the measurement process, the cold-end temperature was fixed at 300 K and the hot-end temperature was set at 364 K. The power generation performances of the self-made Bi_2Te_3-based TE devices including the working voltage V_w, working current I_w, and output power P_{out} are shown in Figure 5. I_w as a function of V_w for all the TE devices exhibits a linear relation, manifesting a typical power generation behavior. The as-assembled device with the Ni-based alloy barrier layer has the highest open-circuit voltage of 2.7 V, and the maximum P_{out} values of all the devices are near 1.25 V. According to the Seebeck effect, the voltage generated at a device temperature difference of ΔT ($\Delta T = T_h - T_c$) is $V = 127^*S_{np}$ ($T_h - T_c$), where the total Seebeck coefficient $S_{np} = S_p - S_n$. Theoretically, the open-circuit voltage for our self-made TEC1-12706 devices should be $V = 127^*S_{np}$ ($T_h - T_c$) = 3.18 V, and the working voltage at which the maximum output power is reached should be 1.59 V. However, the measured open-circuit voltage of the self-made device is 2.70 V and the maximum output power reached is at 1.25 V. We considered that the lowered open-circuit voltage and working voltage should be closely related to the device manufacturing process, especially the leg cutting and soldering processes. There are several issues that appear during the device assembly that could lead to a lowered open-circuit voltage: the nonuniform length of the TE legs, the nonuniform soldering of the TE legs, and the formation of the IRL during the assembly process. The nonuniform length and soldering of TE legs leads to different thermal contacts between the TE legs and ceramic sheets, which finally causes a lowered open-circuit voltage due to the parasitic nonuniform temperature difference. The formation of the IRL will lead to the cancelation of the Seebeck voltage since it has been reported that the p-type IRL was found in the n-type legs.[19] Therefore, the higher open-circuit voltage for our self-made TEC1-12706 device should be attributed to the improved interfacial stability and suppressed IRL formation. As all the devices including the self-made and commercial ones have their maximum output power values near 1.25 V, it could be concluded that the device assembly process of TEC1-12706 is reliable.

The maximum P_{out} of the as-assembled device with the Ni metal barrier layer is only 0.49 W while that of the as-assembled device with the Ni-based alloy barrier layer is as high as 0.55 W and increased by 12.2%. Compared with the commercial devices, the power generation performances of our self-made Bi_2Te_3-based TE devices with the Ni-based alloy barrier layer are also better. Based on the TE properties, the theoretical device power generation efficiency η was predicted to be around 3.4% for the self-made Bi_2Te_3-based TE devices with the Ni-based alloy barrier layer (see Note S1).

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In order to further discern the important role of the Ni-based alloy barrier layer in the device performance, both n-type and p-type Bi_2Te_3-based legs electroplated with the Ni metal barrier layer and Ni-based alloy barrier layer were first annealed at 200 °C for different times and then assembled into two series of Bi_2Te_3-based TE devices. The internal resistance R of the Bi_2Te_3-based TE devices was measured, and the results are shown in Figure 6a. It can be seen that the R value of the two kinds of devices with the Ni metal barrier layer increased dramatically upon increasing the annealing time. The increase in the R value is closely related to the formation of the IRL between the Ni metal barrier layer and Bi_2Te_3-based legs. Upon increasing the annealing time, the IRL with high resistances formed gradually, which led to the increase in the R value of devices. After having been annealed for 6 days, the R value of the as-assembled device with the Ni metal barrier layer increased from 2.10 to 2.75 Ω and increased by 31%, while the R value of these devices with the Ni-based alloy barrier layer increased from 2.02 to 2.51 Ω and increased by only 24%. Therefore, the use of the Ni-based alloy barrier layer can effectively alleviate the severe increase of the internal resistance, as is common in traditional Bi_2Te_3-based TE devices with the Ni metal barrier layer.

The increase in the internal resistance of two kinds of the as-assembled Bi_2Te_3-based TE devices above has significant influence on the cooling performance. It is known that the cooling capacity Q_c describes the amount of heat through hot and cold ends of a TE device, which is defined as

$$Q_c = S_{np}T_cI - \frac{1}{2}I^2R - K(T_h - T_c) \quad (1)$$

where S_{np} is the total absolute Seebeck coefficient of the n-type and p-type TE legs, I is the loop current, K is the total thermal resistance of the TE device, and T_h and T_c are the temperatures at the hot and cold ends of the TE device, respectively. Therefore, the increase in the internal resistance of the two as-assembled Bi_2Te_3-based TE devices may directly contribute to the Joule heat I^2R, resulting in a decrease in Q_c. The maximum Q_c values of the two kinds of the as-assembled Bi_2Te_3-based TE devices are shown in Figure 6b. The maximum Q_c values of the two kinds of the as-assembled Bi_2Te_3-based TE devices with different barrier layers both decrease upon increasing the annealing time. After having been annealed for 6 days, the maximum Q_c of the as-assembled Bi_2Te_3-based devices with the Ni metal barrier layer significantly varied from 53.3 to 40.8 W and decreased by about 24%, while the maximum Q_c of the as-assembled Bi_2Te_3-based devices with the Ni-based alloy barrier layer slightly varied from 55.3 to 47.8 W and only decreased by about 14%. These results clearly indicated that the as-assembled Bi_2Te_3-based devices with the Ni-based alloy barrier layer have better stability in cooling performance. Therefore, it can be expected that both the service stability and service life of the Bi_2Te_3-based TE devices can be largely improved by introducing the Ni-based alloy barrier layer between the Bi_2Te_3-based TE legs and Cu electrodes.

4. CONCLUSIONS

In this work, a high-quality barrier layer of a Ni-based alloy has been fabricated on both n-type and p-type Bi_2Te_3-based TE legs by the electroplating method to improve the thermal stability of the Bi_2Te_3-based TE devices. Unlike the severe chemical reaction between the Ni metal barrier layer and Bi_2Te_3-based TE legs, the Ni-based alloy barrier layer has no significant reaction with the Bi_2Te_3-based TE legs even at 300 °C. Two kinds of Bi_2Te_3-based TE legs with different barrier layers had been annealed at 200 °C for different days to investigate the evolution law of the IRL between the barrier layers and TE legs. The in situ XRD results and high-resolution HAADF-STEM images reveal that the Ni-based alloy barrier layer has more excellent stability than that of the Ni metal barrier layer. The IRL thickness remarkably decreased from 800 to 500 nm for the Ni metal to 100−50 nm for the Ni-based alloy after having been annealed at 200 °C for 48 h. The Bi_2Te_3-based TE devices with excellent structural and performance stabilities were assembled with the as-grown high-performance n-type and p-type Bi_2Te_3-based leg with the Ni-based alloy barrier layer, which have lower internal resistance and higher cooling and power generation performances. A maximum cooling temperature difference over 65 K and a maximum cooling capacity of 55 W were obtained for the high-performance Bi_2Te_3-based TE devices. These results indicate that for the commercial Bi_2Te_3-based TE devices, designing the Ni-based alloys as the barrier layers between the TE legs and electrodes may simultaneously help in achieving excellent performances and thermal stability.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.1c24627.

FESEM images of the Ni metal and Ni-based alloy samples; SAED images of the Ni metal barrier layer and Ni-based barrier layer; HAADF-STEM and elemental mapping at the interfaces after annealing treatment of the Ni metal and Ni-based alloy samples; temperature dependence of the power factor, electronic thermal conductivity and lattice thermal conductivity for n-type and p-type Bi_2Te_3-based crystalline rods prepared in this work and purchased commercial products; electrical transport properties of the as-grown and commercial Bi_2Te_3-based ingots; lattice parameter changes of the p-type and n-type Bi_2Te_3-based wafers electroplated with the Ni-metal and Ni-based alloy barrier layers; and theoretical conversion efficiency calculation (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Ping Wei − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China*; Email: pingwei@whut.edu.cn

Wenyu Zhao − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China*; orcid.org/0000-0002-7468-9790; Email: wyzhao@whut.edu.cn

Authors

Wu Zhu − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China*

Jianqiang Zhang − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China*

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Longzhou Li − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China*
Wanting Zhu − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China;* orcid.org/0000-0002-3067-997X
Xiaolei Nie − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China*
Xiahan Sang − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China;* orcid.org/0000-0002-2861-6814
Qingjie Zhang − *State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, Wuhan University of Technology, Wuhan 430070, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsami.1c24627

Author Contributions

W.Z. and Q.Z. supervised the overall project. W.Z. carried out the experiments and property measurements. P.W. and W.Z. wrote the manuscript. L.L. and J.Z. assembled the devices. B.W. and Y.T. helped to prepare the samples. W.T.Z., X.N., and X.S. performed the material characterization. All authors analyzed the results.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Snyder, G. J.; Toberer, E. S. Complex thermoelectric materials. *Nat. Mater.* **2008**, *7*, 105−114.
(2) Zebarjadi, M.; Esfarjani, K.; Dresselhaus, M. S.; Ren, Z. F.; Chen, G. Perspectives on thermoelectrics: from fundamentals to device applications. *Energy Environ. Sci.* **2012**, *5*, 5147−5162.
(3) Zong, P.-a.; Hanus, R.; Dylla, M.; Tang, Y.; Liao, J.; Zhang, Q.; Snyder, G. J.; Chen, L. Skutterudite with graphene-modified grain-boundary complexion enhances zT enabling high-efficiency thermoelectric device. *Energy Environ. Sci.* **2017**, *10*, 183−191.
(4) Kim, F.; Kwon, B.; Eom, Y.; Lee, J. E.; Park, S.; Jo, S.; Park, S. H.; Kim, B.-S.; Im, H. J.; Lee, M. H.; Min, T. S.; Kim, K. T.; Chae, H. G.; King, W. P.; Son, J. S. 3D printing of shape-conformable thermoelectric materials using all-inorganic Bi_2Te_3-based inks. *Nat. Energy* **2018**, *3*, 301−309.
(5) Li, C.; Ma, S.; Wei, P.; Zhu, W.; Nie, X.; Sang, X.; Sun, Z.; Zhang, Q.; Zhao, W. Magnetism-induced huge enhancement of the room-temperature thermoelectric and cooling performance of p-type BiSbTe alloys. *Energy Environ. Sci.* **2020**, *13*, 535−544.
(6) Zhang, Q.; Zhou, Z.; Dylla, M.; Agne, M. T.; Pei, Y.; Wang, L.; Tang, Y.; Liao, J.; Li, J.; Bai, S.; Jiang, W.; Chen, L.; Snyder, G. J. Realizing high-performance thermoelectric power generation through grain boundary engineering of skutterudite-based nanocomposites. *Nano Energy* **2017**, *41*, 501−510.
(7) Hao, F.; Qiu, P.; Tang, Y.; Bai, S.; Xing, T.; Chu, H.-S.; Zhang, Q.; Lu, P.; Zhang, T.; Ren, D.; Chen, J.; Shi, X.; Chen, L. High efficiency Bi_2Te_3-based materials and devices for thermoelectric power generation between 100 and 300 °C. *Energy Environ. Sci.* **2016**, *9*, 3120−3127.
(8) Hu, L.-P.; Zhu, T.-J.; Wang, Y.-G.; Xie, H.-H.; Xu, Z.-J.; Zhao, X.-B. Shifting up the optimum figure of merit of p-type bismuth telluride-based thermoelectric materials for power generation by suppressing intrinsic conduction. *NPG Asia Mater.* **2014**, *6*, No. e88.
(9) Li, Y.; Qiao, J.; Zhao, Y.; Lan, Q.; Mao, P.; Qiu, J.; Tai, K.; Liu, C.; Cheng, H. A flexible thermoelectric device based on a Bi_2Te_3-carbon nanotube hybrid. *J. Mater. Sci. Technol.* **2020**, *58*, 80−85.
(10) Liu, F.; Wu, Y.-H.; Zhang, Q.; Zhu, T.-J.; Zhao, X.-B. Enhancing room-temperature thermoelectric performance of n-type Bi_2Te_3-based alloys via sulfur alloying. *Rare Met.* **2020**, *40*, 513−520.
(11) Mengali, O. J.; Seiler, M. R. Contact resistance studies on thermoelectric materials. *Adv. Energy Convers.* **1962**, *2*, 59−68.
(12) Lan, Y. C.; Wang, D. Z.; Chen, G.; Ren, Z. F. Diffusion of nickel and tin in p-type $(Bi,Sb)_2Te_3$ and n-type $Bi_2(Te,Se)_3$ thermoelectric materials. *Appl. Phys. Lett.* **2008**, *92*, 101910.
(13) Chen, S.-w.; Chiu, C.-n. Unusual cruciform pattern interfacial reactions in Sn/Te couples. *Scripta Mater.* **2007**, *56*, 97−99.
(14) Chen, S.-w.; Wu, H.-j.; Wu, C.-y.; Chang, C.-f.; Chen, C.-y. Reaction evolution and alternating layer formation in Sn/$(Bi_{0.25}Sb_{0.75})_2Te_3$ and Sn/Sb_2Te_3 couples. *J. Alloys Compd.* **2013**, *553*, 106−112.
(15) Zhou, H.; Mu, X.; Zhao, W.; Tang, D.; Wei, P.; Zhu, W.; Nie, X.; Zhang, Q. Low interface resistance and excellent anti-oxidation of Al/Cu/Ni multilayer thin-film electrodes for Bi_2Te_3-based modules. *Nano Energy* **2017**, *40*, 274−281.
(16) Liu, W.; Bai, S. Thermoelectric interface materials: A perspective to the challenge of thermoelectric power generation module. *J. Mater. Res.* **2019**, *5*, 321−336.
(17) Chu, J.; Huang, J.; Liu, R.; Liao, J.; Xia, X.; Zhang, Q.; Wang, C.; Gu, M.; Bai, S.; Shi, X.; Chen, L. Electrode interface optimization advances conversion efficiency and stability of thermoelectric devices. *Nat. Commun.* **2020**, *11*, 2723.
(18) Wang, X.; Chen, L.; Bai, S.; Shi, X.; Song, Q.; Liao, J.; Gu, M. High temperature interfacial stability of Fe/$Bi_{0.5}Sb_{1.5}Te_3$ thermoelectric elements. *J. Materiomics* **2021**, *36*, 197.
(19) Liu, W.; Wang, H.; Wang, L.; Wang, X.; Joshi, G.; Chen, G.; Ren, Z. Understanding of the contact of nanostructured thermoelectric n-type $Bi_2Te_{2.7}Se_{0.3}$ legs for power generation applications. *J. Mater. Chem. A* **2013**, *1*, 13093−13100.
(20) Ren, Z.; Lan, Y.; Zhang, Q. *Advanced Thermoelectrics, Materials, Contacts, Devices, and Systems*; CRC Press: New York, 2017; pp 605−625.
(21) Mu, X.; Zhu, W.; Zhao, W.; Zhou, H.; Zhang, Q. J. N. E. Excellent transverse power generation and cooling performances of artificially tilted thermoelectric film devices. *Nano Energy* **2019**, *66*, 104145.
(22) Xiong, K.; Wang, W.; Alshareef, H. N.; Gupta, R. P.; White, J. B.; Gnade, B. E.; Cho, K. Electronic structures and stability of Ni/Bi_2Te_3 and Co/Bi_2Te_3 interfaces. *J. Phys. D: Appl. Phys.* **2010**, *43*, 115303.